# 1009. Complement of Base 10 Integer

Solved

Easy Topics Companies Hint

The **complement** of an integer is the integer you get when you flip all the `0`'s to `1`'s and all the `1`'s to `0`'s in its binary representation.

- For example, The integer `5` is `"101"` in binary and its **complement** is `"010"` which is the integer `2`.

Given an integer `n`, return *its complement*.

**Example 1:**

```
Input: n = 5
Output: 2
Explanation: 5 is "101" in binary, with complement "010" in binary, which is 2 in base-10.
```

**Example 2:**

```
Input: n = 7
Output: 0
Explanation: 7 is "111" in binary, with complement "000" in binary, which is 0 in base-10.
```

**Example 3:**

```
Input: n = 10
Output: 5
Explanation: 10 is "1010" in binary, with complement "0101" in binary, which is 5 in base-10.
```

**Constraints:**

- $0 <= n < 10^9$

# Python:

```
class Solution:
    def bitwiseComplement(self, n: int) -> int:
        if n == 0: return 1
        return ~n & (1 << n.bit_length()) - 1
```

# JavaScript:

```
const bitwiseComplement = n => {
    if (n === 0) return 1;
    return ~n & (1 << 32 - Math.clz32(n)) - 1;
};
```

# Java:

```
class Solution {
    public int bitwiseComplement(int n) {
        if (n == 0) return 1;
        int bits = 32 - Integer.numberOfLeadingZeros(n);
        int mask = (1 << bits) - 1;
        return ~n & mask;
    }
}
```